

EXPERIMENT - 1

Data Acquisition and Data Profiling using BI Tools

Report



Subject Name: Business Intelligence & Tools
Subject Code: 24CSH-341

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1. Aim

To acquire data from multiple sources and perform data profiling using Microsoft Power BI in order to identify data-quality issues and prepare the data for further analysis.

2. Objectives

The objectives of this experiment are:

- To understand the process of data acquisition using BI tools.
- To connect Power BI with different data sources such as CSV, Excel, and databases.
- To import and inspect data using Power Query Editor.
- To understand the concept of data profiling.
- To identify missing, invalid, inconsistent, and unusual data values.
- To analyze column quality, distribution, and profile information.
- To identify suitable data-cleaning and transformation requirements.

3. Software and Tools Required

- Microsoft Power BI Desktop
- Power Query Editor
- CSV dataset
- Microsoft Excel dataset
- Database containing transaction data
- Windows operating system

4. Introduction

Business Intelligence systems collect and analyze data from multiple sources to support data-driven decision-making. However, raw data may contain missing values, incorrect data types, duplicate records, inconsistent entries, and unusual values. Data acquisition is the process of collecting or importing data from different sources into a BI environment.

Data profiling is the process of examining a dataset to understand its structure, completeness, quality, distribution, and possible errors.

In this experiment, Microsoft Power BI and Power Query are used to acquire data from CSV, Excel, and database sources. The imported data is then profiled using Power Query's Column Quality, Column Distribution, and Column Profile features.

5. Data Sources

5.1 Sales Data – CSV

The Sales Data CSV file contains sales-related information such as Customer ID, Product, Quantity, Price, Date, and Sales Amount. The dataset was imported into Power BI using the Text/CSV connector.

5.2 Employee Database – Excel

The Employee Database contains information about employees, including Employee ID, Employee Name, Department, Joining Date, Salary, and other employee-related attributes. The Excel file was imported using the Excel Workbook connector.

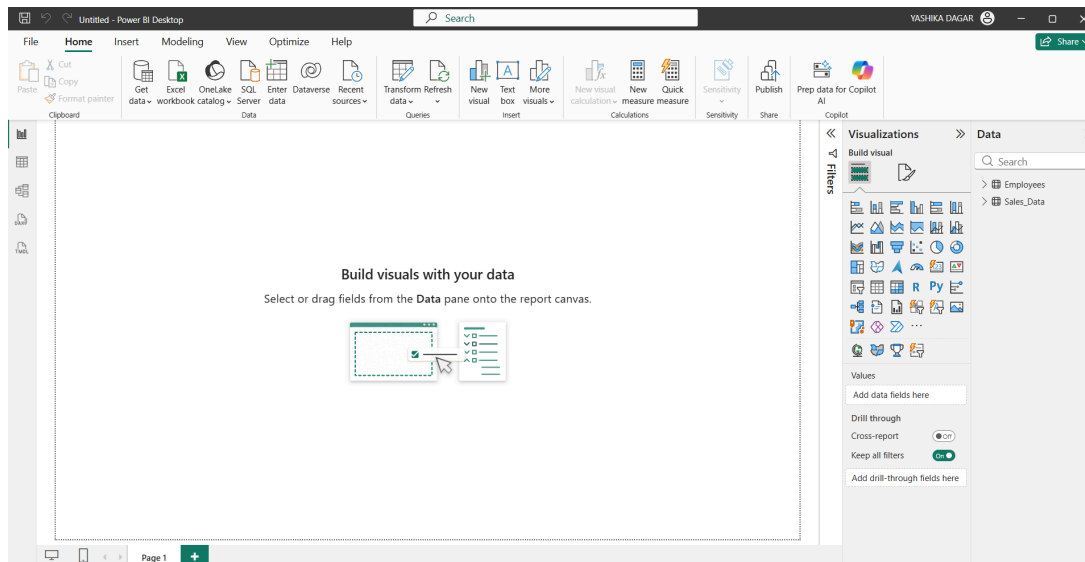
5.3 Transaction Data – Database

Transaction data was obtained from a database using the appropriate database connector in Power BI. The transaction dataset contains fields related to Transaction ID, Customer ID, Transaction Date, Transaction Amount, and Transaction Type.

6. Procedure

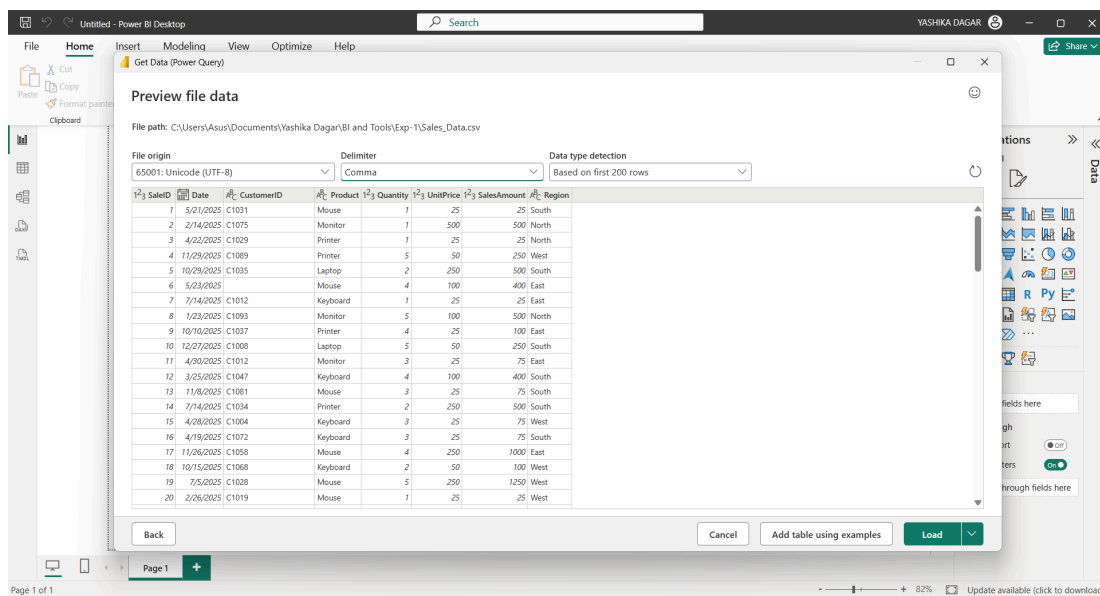
Step 1: Launch Power BI

Microsoft Power BI Desktop was opened to begin the data-acquisition process.



Step 2: Import Sales Data

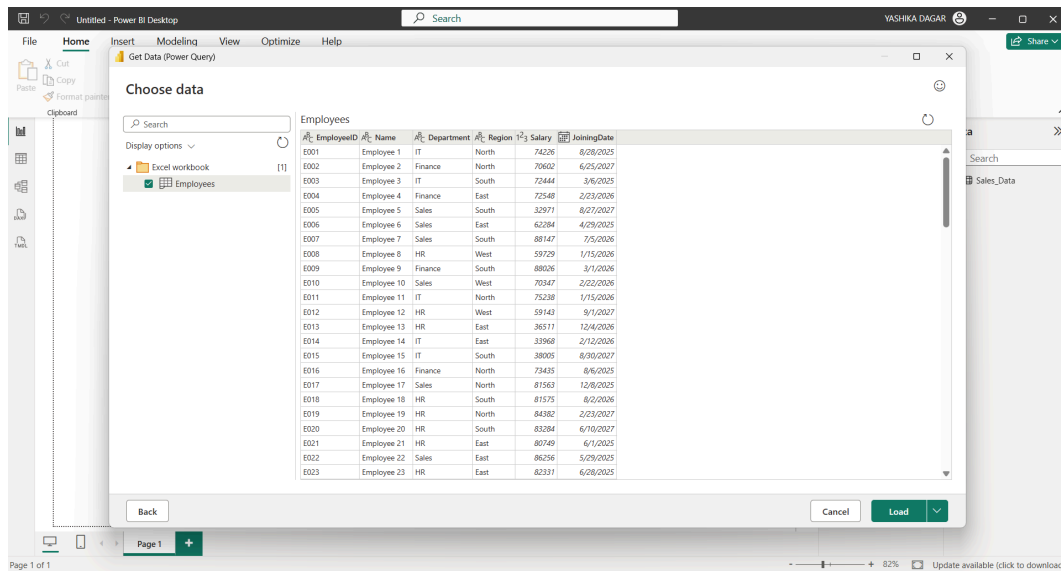
The Get Data → Text/CSV option was selected and the Sales Data CSV file was loaded into Power BI.



Step 3: Import Employee Data

The Get Data → Excel Workbook option was selected and the Employee Database Excel file was imported.

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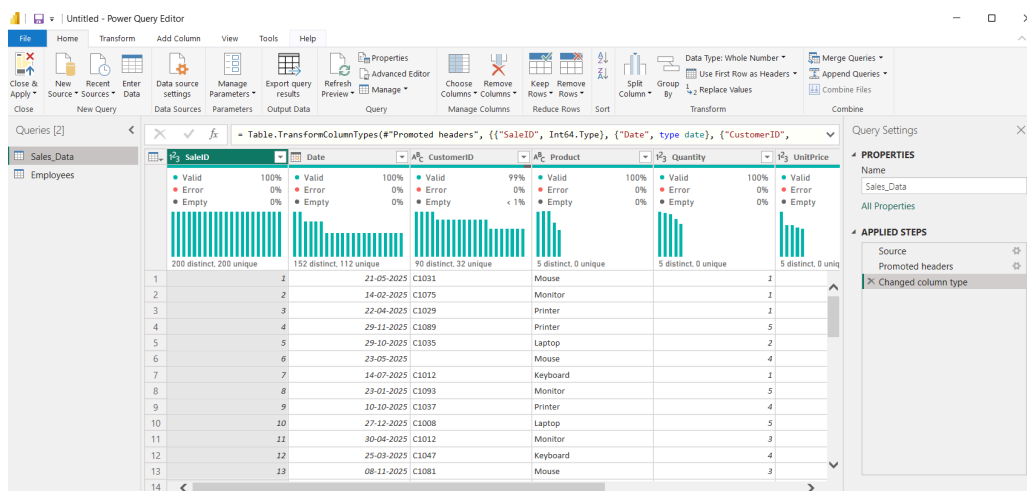


Step 4: Connect to Transaction Database

The appropriate database connector was selected and the required transaction table was loaded into Power BI.

Step 5: Open Power Query Editor

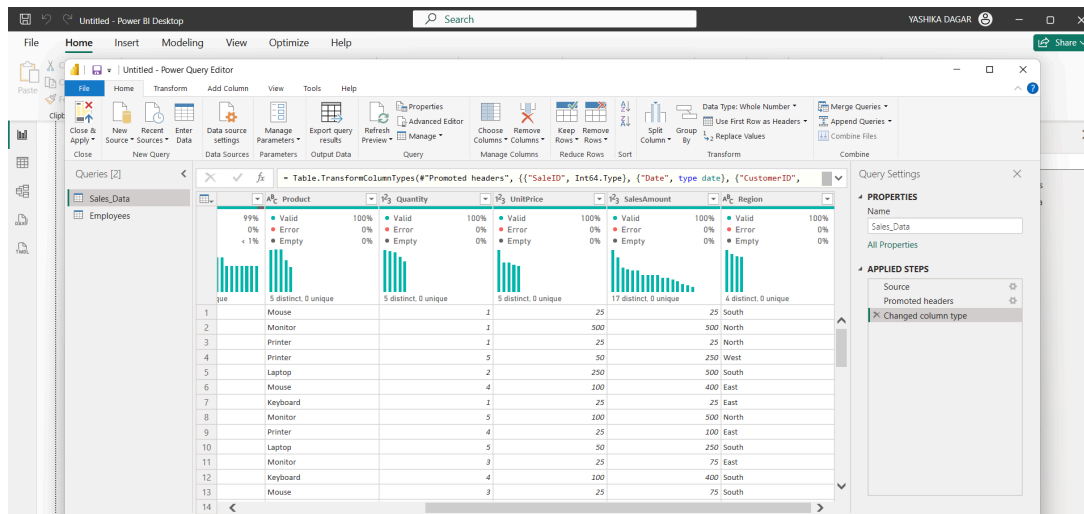
The Transform Data option was selected to open Power Query Editor.



Step 6: Inspect the Data

The imported tables were examined to understand their columns, values, data types, and overall structure.

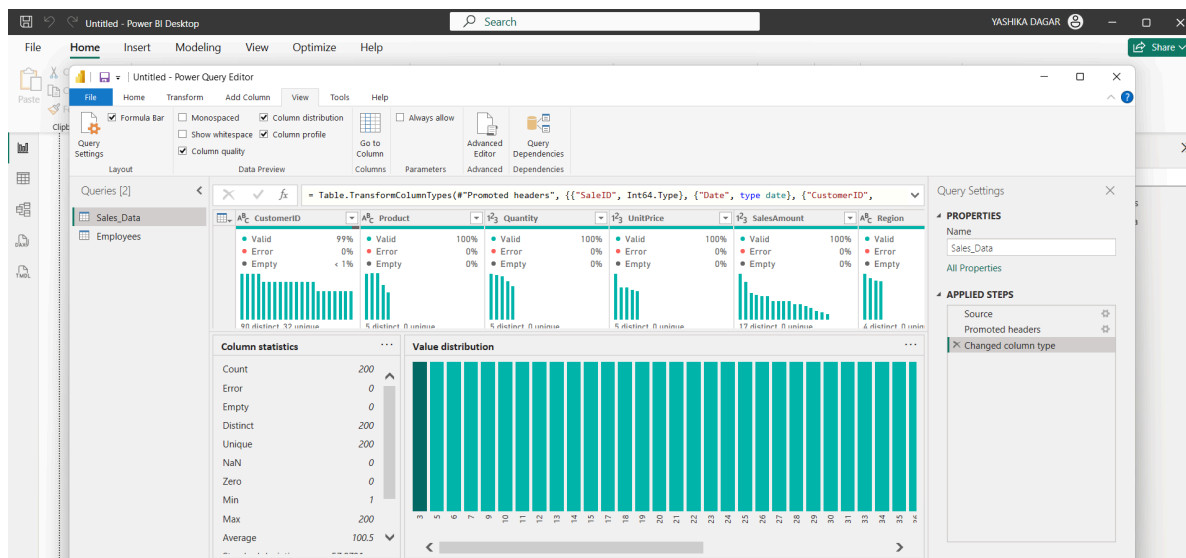
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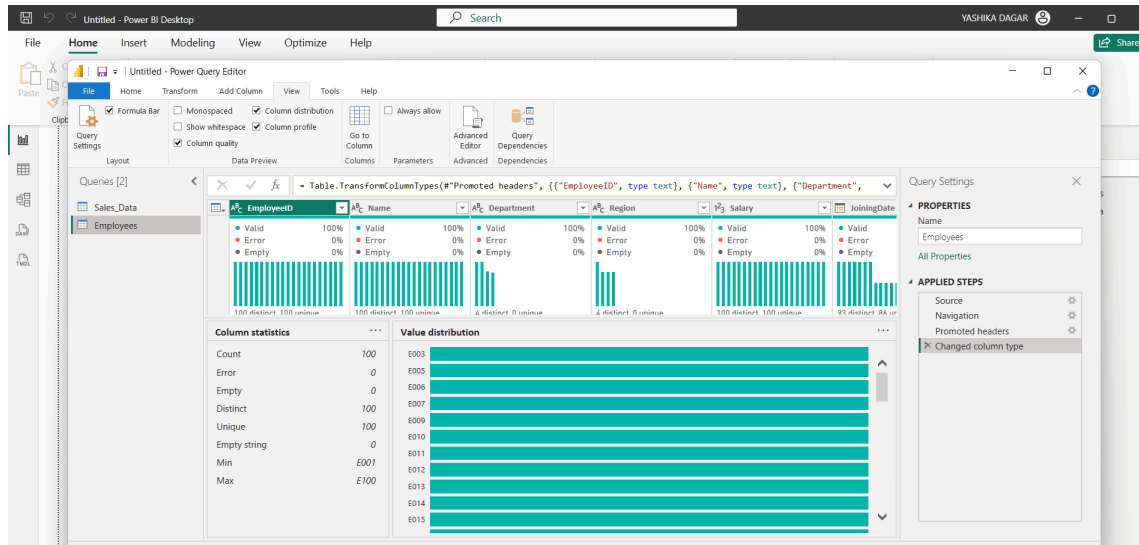


Step 7: Enable Data Profiling

The following options were enabled from the View tab:

- Column Quality
- Column Distribution
- Column Profile

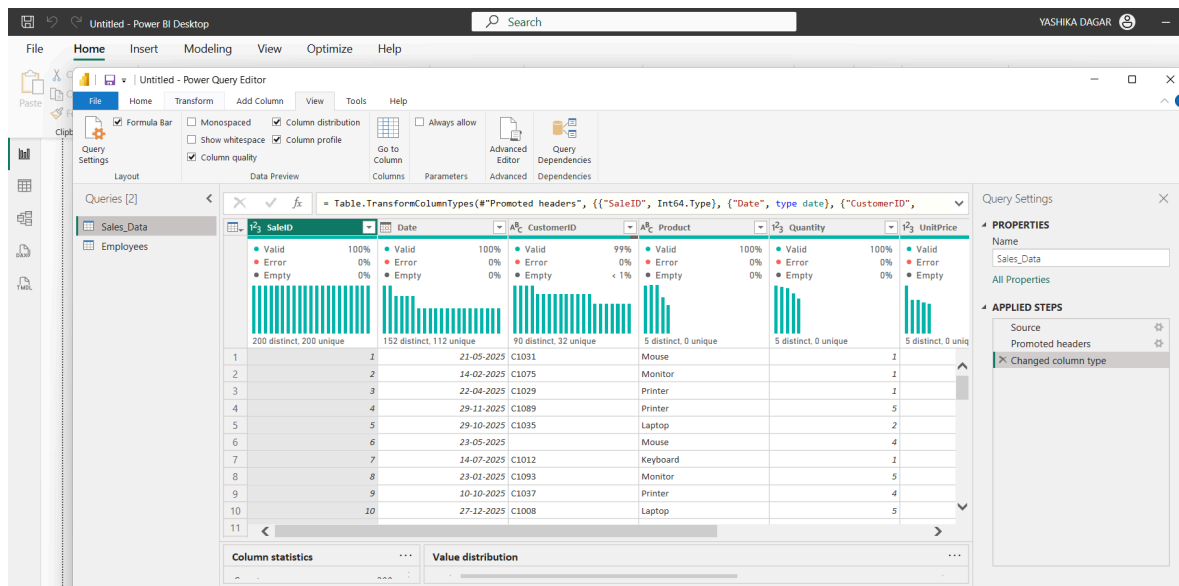




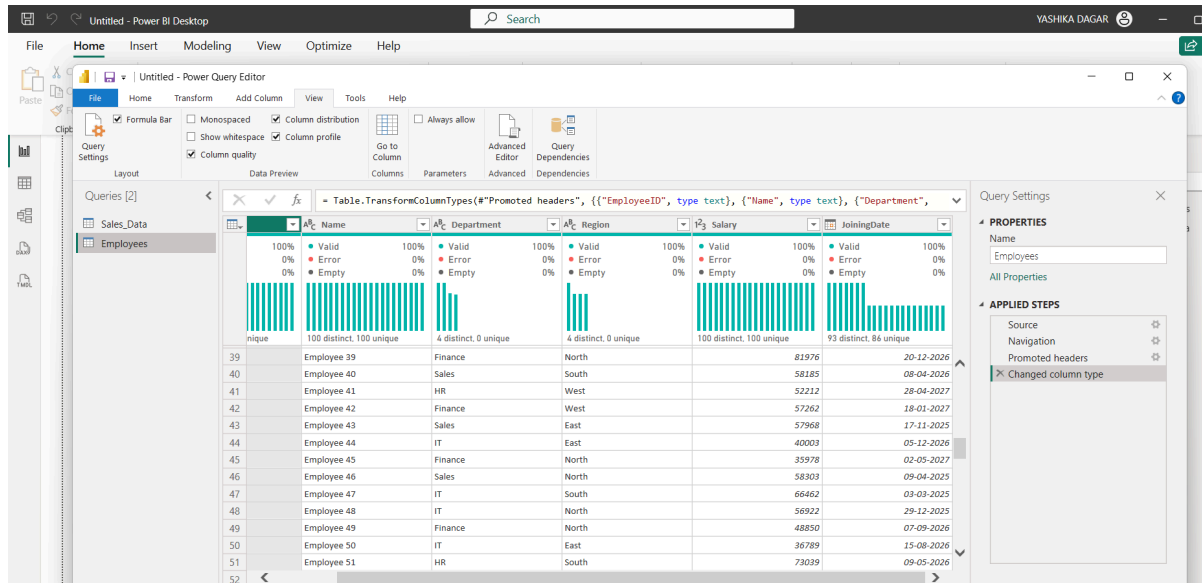
Step 8: Analyze Data Quality

Each column was inspected for:

- Valid values
- Empty/null values
- Errors
- Unusual values
- Incorrect data types
- Inconsistent entries



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7. Data Profiling

7.1 Column Quality

Column Quality displays the percentage of valid, empty, and error values in a column.

7.2 Column Distribution

Column Distribution provides information about the distribution of values in a column. It helps identify distinct, unique, and frequently occurring values.

7.3 Column Profile

Column Profile provides detailed statistics for the selected column, including the number of rows, valid values, empty values, errors, and other statistics depending on the data type.

8. Observations

8.1 Sales Data

The Sales Data was examined using Column Quality and Column Profile.

One of the identified issues was the presence of missing Customer ID values.

8.2 Employee Database

The Employee Database was examined for valid values, data types, and unusual records.

An unusual observation was found in the Joining Date field, where some dates appeared to be future dates.

8.3 Transaction Data

The transaction dataset was profiled to examine transaction dates, amounts, and other attributes.

An unusual transaction date such as 01-01-2026 was identified during the data-quality examination.

Results

The experiment successfully demonstrated the acquisition and profiling of data using Microsoft Power BI.

The following results were obtained:

- Data was successfully acquired from different sources.
- CSV, Excel, and database data were loaded into Power BI.
- Power Query Editor was used for data inspection.
- Column Quality, Column Distribution, and Column Profile were used for data profiling.
- Missing and unusual values were identified.
- Data-quality issues were documented.
- Appropriate data-cleaning actions were suggested.
- The datasets were prepared for further analysis and visualization.

Learning Outcomes

After completing this experiment, I was able to:

- Understand the importance of data acquisition in Business Intelligence.
- Import data from CSV, Excel, and database sources into Power BI.
- Use Power Query Editor for inspecting and preparing data.
- Perform data profiling using Column Quality, Column Distribution, and Column Profile.
- Identify missing, invalid, and unusual values.
- Analyze basic data-quality issues.

- Suggest appropriate data-cleaning and transformation methods.
- Understand the importance of data quality before performing BI analysis.

Conclusion

The experiment provided practical knowledge of data acquisition and data profiling using Microsoft Power BI. Data from multiple sources was successfully imported and examined using Power Query.

The profiling process helped identify issues such as missing Customer IDs, unusual joining dates, and unusual transaction dates. These observations demonstrate that data profiling is an important step before performing analysis because the quality of the input data directly affects the reliability of the resulting analysis.

Thus, the experiment successfully demonstrated how BI tools can be used to acquire, inspect, profile, and prepare data for further analysis and decision-making.